MEDSTOCK AI

MedicineSupplyIntelligence&PublicAccountability System

Every Medicine Accounted For.
Every Patient Informed.
An AI-powered platform to improve medicine traceability, predict shortages and strengthen public accountability.

The idea in one minute

Governmenthospitals need reliable medicinerecords so that essential medicines reach patients on time. Digital inventory systems already exist, but records can be delayed, incomplete or difficult to compare across suppliers, warehouses and hospitals. Patients may also be unable to verify whether a medicine is genuinely available.

MedStock AI adds an intelligence and accountability layer. It compares supplier dispatches, hospital receipts and recorded distribution; flags unusual stock patterns for verification; predicts possible shortages; recommends safe transfers of surplus stock; and lets citizens report medicine non-availability.

What makes it different

TRACE	PREDICT	OPTIMISE	ACCOUNT
Track medicine batches and movements.	Warn about likely stockouts and expiry.	Recommend transfers between facilities.	Give citizens a verified reporting route.

Core principle: A data mismatch is a signal to investigate, not proof of theft or corruption. Human verification and fair process remain essential.

Proposed scope: accountability-first, supported by AI anomaly detection, shortage forecasting and medicine redistribution.

1. The problem and the opportunity

Medicine shortages andunreliablestock information can have severalcauses: procurement delays, incomplete records, damaged or expired stock, distribution errors, unexpected demand and, in some cases, possible diversion. A useful solution must help distinguish between these causes instead of assuming every shortage is fraud.

Why an additional layer is worth exploring

Existing challenge	Why it matters	MedStock AI response
Dispatch and receipt figures differ.	The facility may not have received the recorded quantity.	Compare dispatch, receipt and acknowledgement records.
Stock records are not updated on time.	Patients and officials may see inaccurate availability.	Show last update time and flag stale records.
Shortages are noticed late.	Patients may visit after stock has run out.	Forecast stockout dates using consumption history.
Surplus stock is not easily shared.	One facility may have excess while another faces a shortage.	Recommend approved transfers after checking expiry and reserves.
Citizens cannot easily report a mismatch.	Repeated problems may remain hidden.	Provide complaint IDs, status tracking and official verification.

Existing systems and the gap

Government platforms such as e-Aushadhi already support medicine procurement and inventory functions. Haryana also has medicine procurement and inventory infrastructure. MedStock AI should therefore be presented as a complementary intelligence and accountability layer, not as the first medicine inventory system.

Audit findings have highlighted weaknesses in implementation and record quality in particular settings. For example, CAG audits have reported incomplete or delayed inventory updates, receipt/distribution record gaps and weak audit trails in specific state or facility contexts. These are historical, location-specific findings—not proof that every hospital or current system is failing.

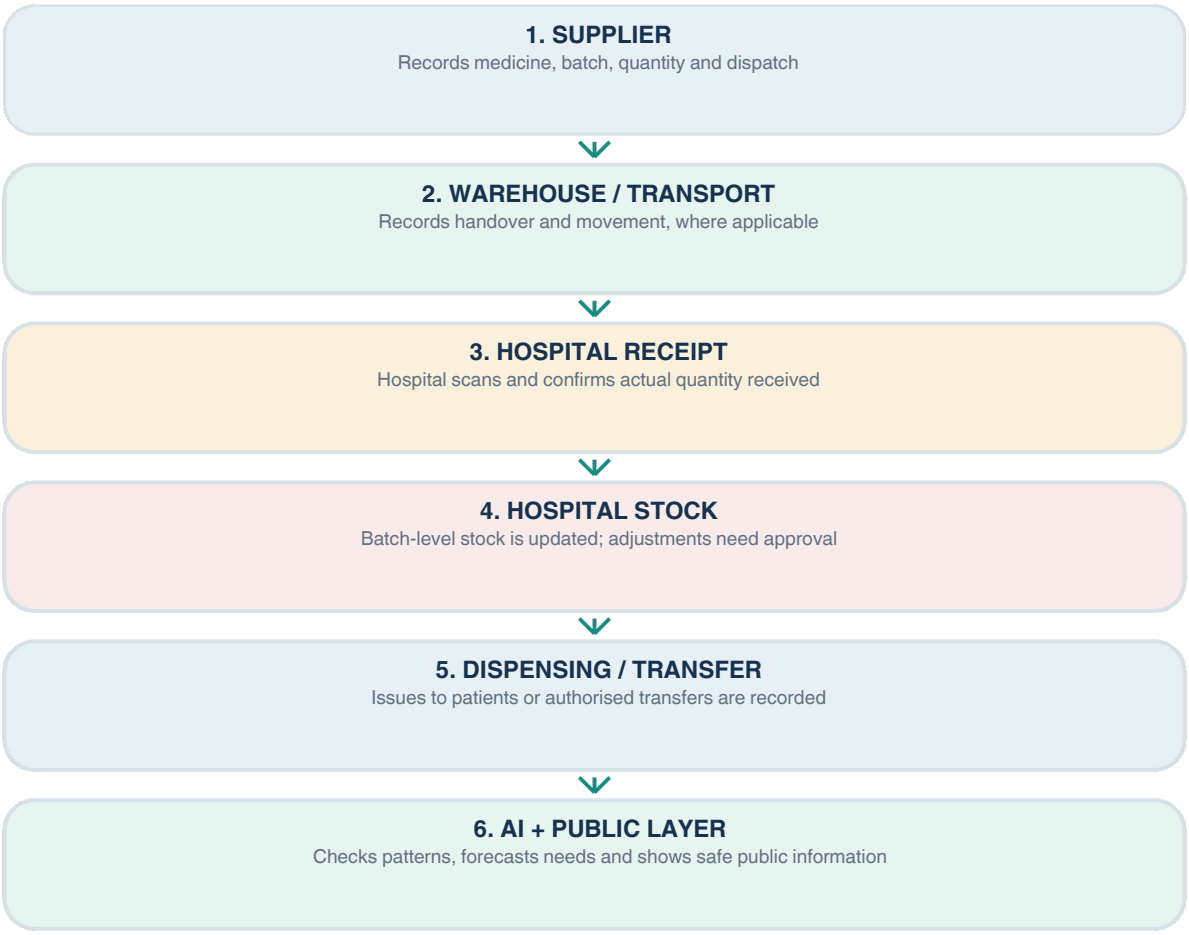
The gap we propose to test

Can independent reconciliation, AI risk signals, stock forecasting and citizen reports make existing medicine data more reliable and actionable?

Before claiming a national problem or a new capability, validate the gap with hospital pharmacists, procurement staff and patients. Existing grievance portals may already accept medicine-availability complaints; MedStock AI should complement or connect to those channels.

2. How the system works

MedStockAI followsamedicinefromdispatchto thehospital and records the stock movement that follows. Separate confirmations are important: a supplier's dispatch entry alone does not prove that the hospital received the full quantity.

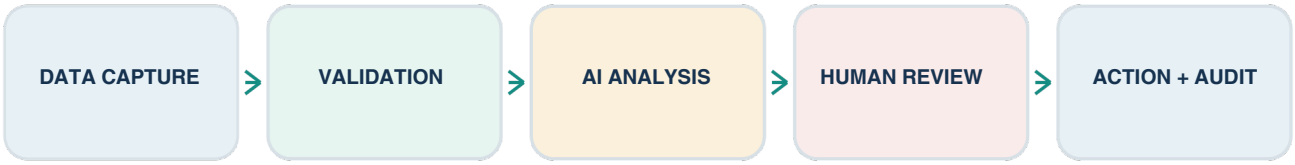


Stock reconciliation

Recorded closing stock = opening stock + verified receipts + approved transfers in – recorded issues – approved transfers out – authorised adjustments.

The system compares expected stock with physical counts during scheduled checks. A difference creates a reconciliation task. It does not automatically label the difference as theft.

Data and verification flow



Key safeguards

- Eachtransaction hasa unique ID, timestamp, user role and supporting record.
- Corrections are recorded as new authorised entries; old entries are not silently overwritten.
- QR codes identify medicine batches, but scanning alone does not prove physical quantity.
- Public stock information shows when it was last updated and whether it has been physically verified.

3. Main features: supply-chain intelligence

A. End-to-end medicine traceability

Supplier and hospital usersrecorddispatches, receipts, transfers and distribution. Batch number, expiry date, medicine strength and quantity are stored. QR scanning can make data entry faster and reduce typing errors.

Example: a supplier records 1,000 units dispatched, but the hospital confirms receipt of 980. MedStock AI flags the 20-unit difference and asks for a reason such as partial delivery, damage or a data-entry error.

B. AI anomaly and risk detection

Thesystem looksfor patternsthat areunusualcompared with normal stock movement. Examples include repeated unexplained adjustments, unusual wastage, large quantity differences or stock shown as available despite repeated reports that it cannot be obtained.

Signal	What the system does
Dispatch and receipt mismatch	Creates a reconciliation alert and requests supporting records.
Repeated unusual wastage	Highlights a pattern for audit review.
Sudden unexplained stock reduction	Checks transaction history and requests verification.
Repeated citizen reports	Groups similar reports and compares them with stock data.

Possible AI method: Isolation Forest or another anomaly-detection model, supported by clear rule-based checks. A risk score prioritises review; it is not a finding of guilt. Human investigation is required.

C. Shortage and expiry prediction

Thesystemuses past consumption, currentusable stock, pending deliveries and seasonal demand to estimate when a medicine may run out. It can also flag batches approaching expiry so that authorised staff can review safe use or transfer options.

Simple stock-cover example: 500 tablets ÷ 80 tablets used per day = 6.25 days of stock. If the next delivery is expected in 10 days, the system warns that a shortage may occur before delivery.

Potential tools: Python, Pandas and Scikit-learn; XGBoost or time-series forecasting can be explored if enough historical data is available. Start with a transparent baseline model and compare its predictions with actual outcomes.

4. Redistribution and public accountability

D. Smart medicine redistribution

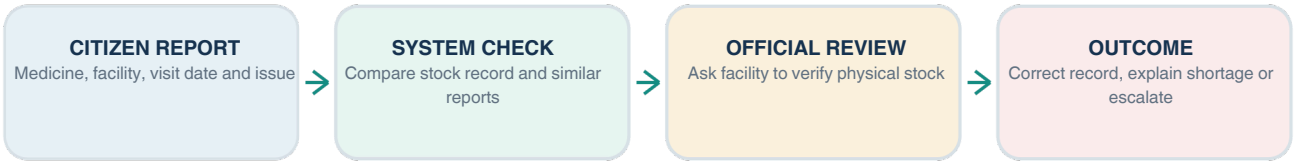
Whenone hospitalhas more usable stockthan it is likely to need and another is at risk of running out, MedStock AI recommends a possible transfer. The recommendation checks expiry, storage conditions, minimum reserve stock, medicine rules, distance and transport time. An authorised officer approves the transfer.

Hospital	Usable stock	Estimated need	Possible insight
Hospital A	1,000	300	May have transferable surplus.
Hospital B	100	500	Possible shortage risk.
Hospital C	400	350	May need to retain reserve.

E. Public medicine availability portal

Citizens select a district, hospitaland medicineto see the facility's reported availability, the last update time and whether the information has been physically verified. The portal must not promise that a medicine will definitely be dispensed: eligibility, prescription, stock changes and hospital rules may affect dispensing.

F. Citizen reporting and official investigation



A complaint receives a reference number and status. Personal details are protected. Public users see only appropriate status and verified outcomes. A report is an allegation to check—not proof of wrongdoing. Repeated or substantiated discrepancies can be escalated through the proper government process.

Public dashboard example (illustrative only)

Medicine	Recorded stock	Status shown	Update
Paracetamol 500 mg			
Amoxicillin 500 mg	12	Low recorded stock	10:30 AM
ORS sachets	0	Recorded unavailable	10:30 AM

Never publish patient names, prescriptions, individual dispensing histories, sensitive security information or unverified accusations.

6. Demo, impact and future rollout

Three-minute demonstration story

Demo	What happens	What judges see
1. Discrepancy	Supplier dispatches 1,000 units; hospital confirms 800.	A 200-unit mismatch is flagged for verification.
2. Shortage	Hospital B is forecast to run out in four days; Hospital A has usable surplus.	AI recommends a transfer subject to approval.
3. Citizen report	Portal shows stock, but a patient reports non-availability.	A complaint is logged and sent for official checking.

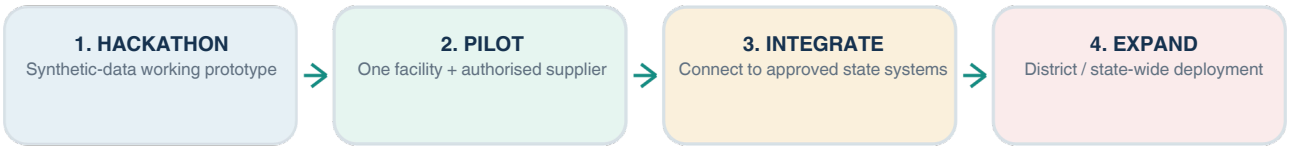
How we measure success

Metric	Prototype test
Discrepancy detection	Share of planted stock mismatches correctly flagged.
False alerts	Share of alerts that are not genuine anomalies in the test set.
Forecast quality	Difference between predicted and actual stockout dates.
Redistribution	Number of simulated shortage risks addressed by feasible transfer plans.
Complaint workflow	Time to assign, verify and close a test complaint.

Safety, fairness and trust

- AI alertsarerisk signals,neverautomatic accusations or proof of corruption.
- An authorised human checks physical stock, supporting records and explanations before action.
- Use role-based access, secure authentication, encryption and an audit trail.
- Protect patient and complainant data; publish only appropriate aggregate stock information.
- Show last-updated time and distinguish recorded stock from physically verified stock.

Future rollout



60-second pitch

Imagine visiting agovernment hospital for a medicine that is supposed to be available, only to be told it is out of stock. Where did the medicine go? Was it never delivered, was the record not updated, or is there another problem? MedStock AI adds an intelligence and accountability layer to existing medicine systems. It tracks dispatch and receipt, detects unusual stock patterns, predicts shortages, recommends transfers and lets citizens report medicine non-availability. Reports are verified by authorised officials, not treated as automatic proof of wrongdoing. Our goal is simple: every medicine should be accounted for, every shortage should be identified and every patient should have reliable information.

Project positioning: Complement existing government systems. Do not claim a live government integration, proven fraud prevention or real-world AI accuracy until these have been independently tested.